A Pn Heterojunction Diode in Thermal Equilibrium:

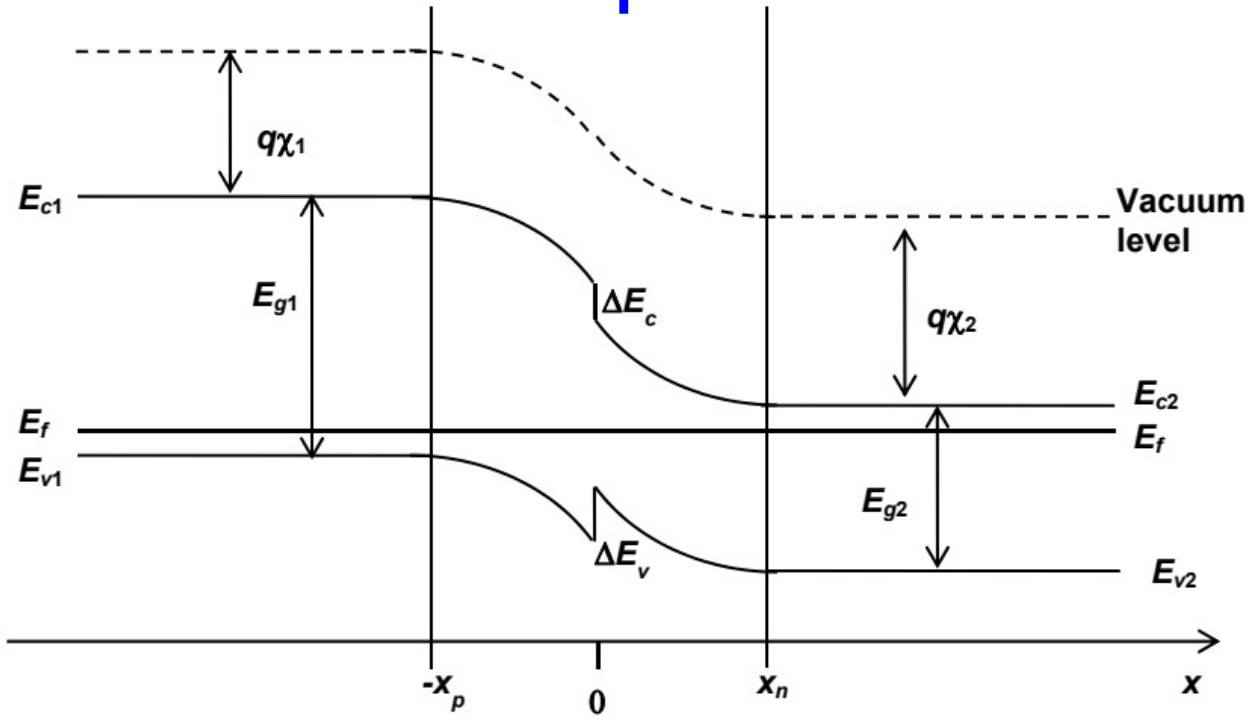
➢ Built-in potential V_{bi}:

$$qV_{bi} = |E_{F1} - E_{F2}|$$

$$qV_{bi} = \Phi_{p1} - \Phi_{n2}$$

$$\Phi_{p1} = q\chi_1 + (E_{C1} - E_{F1})$$
$$= q\chi_1 + E_{g1} - (E_{F1} - E_{V1})$$

$$\Phi_{n2} = q\chi_2 + (E_{C2} - E_{F2})$$



$$qV_{bi} = (q\chi_1 - q\chi_2) + E_{g1} - (E_{F1} - E_{V1}) - (E_{C2} - E_{F2}) \qquad E_{C2} - E_{F2} = -k_B T \ln (N_{D2} / N_{C2})$$

$$V_{bi} = (\chi_1 - \chi_2) + \frac{E_{g1}}{q} - \frac{kT}{q} \ln\left(\frac{N_{V1}}{N_{A1}}\right) - \frac{kT}{q} \ln\left(\frac{N_{C2}}{N_{D2}}\right) \qquad E_{F1} - E_{V1} = -k_B T \ln (N_{A1} / N_{V1})$$